



8-MAVZU · I BOB. HTML VA CSS

Box model va o'lcham birliklari. Flexbox va Grid

Har bir element — quti: content, padding, border, margin; box-sizing; px, %, em, rem, vw, vh; display; Flexbox bilan bir o'lchamli va CSS Grid bilan ikki o'lchamli sahifa maketlari.

Bugun nimani o'rganamiz?

- **Box model:** har bir element — quti
- `margin`, `padding`, `border`, `box-sizing`
- O'lcham birliklari: `px` `%` `em` `rem` `vw` `vh`
- **Flexbox:** menyu, kartalar qatori, markazlash
- **CSS Grid:** sahifa maketi va galereya

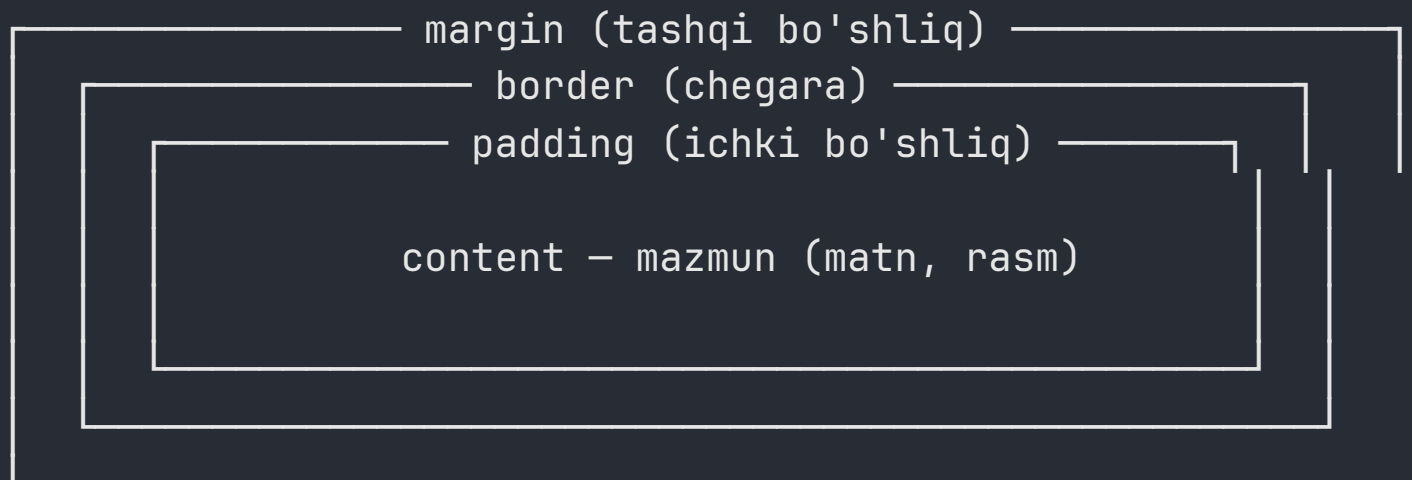
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Box model

Brauzer har bir elementni to'rtburchak quti sifatida chizadi

Qutining 4 qatlami

Matn



Rasmli ramka: content — rasm, padding — oq hoshiya, border — yog'och ramka, margin — devordagi qo'shni ramkagacha masofa.

padding va margin yozish usullari

```
CSS
padding: 20px;           /* 4 tomon */
padding: 10px 20px;      /* tepa-past | chap-o'ng */
padding: 10px 20px 30px; /* tepa | chap-o'ng | past */
padding: 10px 20px 30px 40px; /* tepa o'ng past chap */
padding-top: 10px;       /* bitta tomon */

margin: 0 auto;          /* blokni GORIZONTAL markazlash */
```

- **padding** — ichki bo'shliq, **fon rangi** shu yerga ham tarqaladi
- **margin** — tashqi bo'shliq, **doim shaffof**
- Markazlash: `width: 600px; margin: 0 auto;`

box-sizing — eng muhim bitta qator

- `width: 250px` + `padding: 20px` + `border: 5px` = aslida **300px** 😞

CSS

```
/* Har bir loyiha uchun CSS faylini shu bilan boshlang */  
* {  
  box-sizing: border-box;  
}
```

**border-box: width = content + padding + border → 250px
haqiqatan 250px**

display xossasi

Qiymat	Xatti-harakat
<code>block</code>	Yangi qatordan, butun kenglik; <code>width/height</code> ishlaydi
<code>inline</code>	Qatorda; <code>width/height</code> ishlamaydi
<code>inline-block</code>	Qatorda turadi, lekin <code>width/height</code> ishlaydi
<code>none</code>	Butunlay yashirin, joy egallamaydi
<code>flex</code> / <code>grid</code>	Zamonaviy maket tizimlari

O'lcham birliklari

Birlik	Nimaga nisbatan	Qachon
px	Ekran pikseli	Chegara, soya
%	Ota element	Kengliklar
em	Joriy shrift o'lchami	Tugma padding'i
rem	Sahifa shrifti (16px)	Shrift va bo'shliqlar
vw	Ekran kengligining 1%	Ekran kengligidagi blok
vh	Ekran balandligining 1%	<code>min-height: 100vh</code> hero

- $1.5\text{rem} = 16 \times 1.5 = \mathbf{24\text{px}}$; ota 20px bo'lsa $2\text{em} = \mathbf{40\text{px}}$

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Flexbox

Elementlarni bir yo'nalishda — qator yoki ustunga terish

Flexbox: bitta qator hammasini o'zgartiradi

CSS

```
.konteyner {  
  display: flex;      /* ← farzandlar bir qatorga tiziladi */  
  gap: 10px;          /* oralaridagi masofa */  
}
```

- **Flex konteyner** — `display: flex` berilgan **ota**
- **Flex elementlar** — uning bevosita farzandlari
- Menu, tugmalar qatori, kartalar, markazlash — Flexbox'ning ishi

Ikki o'q

Matn

asosiy o'q (main axis) — justify-content



kesishgan o'q
(cross axis)

▼ align-items



Avtobus bekati: navbat yo'nalishi — asosiy o'q; `justify-content` odamlarni shu chiziq bo'ylab taqsimlaydi. `align-items` — ularni bo'yi bo'yicha tekislaydi.

justify-content va align-items

Xossa	Qiymatlar
<code>justify-content</code>	<code>flex-start</code> <code>center</code> <code>flex-end</code> <code>space-between</code> <code>space-around</code> <code>space-evenly</code>
<code>align-items</code>	<code>stretch</code> (standart) <code>flex-start</code> <code>center</code> <code>flex-end</code>
<code>flex-direction</code>	<code>row</code> <code>column</code> <code>row-reverse</code>
<code>flex-wrap</code>	<code>nowrap</code> <code>wrap</code> — sig'masa keyingi qatorga
<code>gap</code>	<code>16px</code>

CSS

```
/* ✨ To'liq markazlash */  
.k { display: flex; justify-content: center; align-items: center; }
```



`flex-direction: column` da o'qlar almashadi: `justify-content` vertikal ishlaydi

Element xossasi: flex

CSS

```
.k      { display: flex; gap: 8px; }  
.yon    { flex: 0 0 150px; }      /* qat'iy 150px */  
.asosiy { flex: 1; }              /* qolgan BARCHA joy */  
.ikki   { flex: 2; }              /* flex:1 dan 2 barobar keng */
```

Natija

└─yon 150px─┘ └────────── asosiy (flex: 1) ─────────┘
└ flex:1 ┘ └──────── flex:2 ─────────┘ └ flex:1 ┘

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CSS Grid

Qator va ustunlardan iborat to'r — ikki o'lchamli maket

Grid asoslari

CSS

```
.grid {  
  display: grid;  
  grid-template-columns: 1fr 1fr 1fr; /* 3 ta teng ustun */  
  gap: 12px;  
}
```

- **fr** — bo'sh joy **ulushi**: **1fr 2fr** — ikkinchisi 2 barobar keng
- **repeat(4, 1fr)** = **1fr 1fr 1fr 1fr**
- **250px 1fr** — qat'iy yon panel + qolgani



Shaxmat taxtasi / Excel: avval nechta ustun va qatorligini belgilaysiz, keyin elementlarni kataklarga joylaysiz.

✨ Sehrli qator: moslashuvchi galereya

CSS

```
.galereya {  
  display: grid;  
  grid-template-columns: repeat(auto-fill, minmax(150px, 1fr));  
  gap: 10px;  
}  
.galereya img { width: 100%; height: 120px; object-fit: cover; }
```

- "Har ustun **kamida 150px**, nechta sig'sa shuncha, ortgan joyni teng taqsimla"
- Telefonda 2, planshetda 4, kompyuterda 6 ustun — **qo'shimcha kodsiz**

Elementni bir nechta katakka cho'zish

CSS

```
.g { display: grid; grid-template-columns: repeat(4, 1fr); gap: 10px; }  
.katta { grid-column: span 2; grid-row: span 2; } /* 2x2 */  
.keng { grid-column: 1 / 5; } /* 1-chiziqdan 5-chiziqqacha */
```

Natija

2x2 katta	b	c
	d	e
butun qator		

Nomlangan hududlar — sahifa maketi

css

```
.sahifa {  
  display: grid;  
  grid-template-columns: 200px 1fr;  
  grid-template-rows: 70px 1fr 50px;  
  grid-template-areas:  
    "header header"  
    "aside main"  
    "footer footer";  
  min-height: 100vh;  
}  
header { grid-area: header; }  
aside { grid-area: aside; }  
main { grid-area: main; }  
footer { grid-area: footer; }
```

- Maketni **so'zlar bilan chizish**
- Har bir qo'shtirnoq — bitta qator
- Har bir so'z — bitta ustun

Flexbox yoki Grid?

	Flexbox	Grid
O'lcham	Bir o'lchamli	Ikki o'lchamli
Eng yaxshi joyi	Menyu, tugmalar, kartaning ichi, markazlash	Sahifa maketi, galereya, katalog

Raqobatchi emas: skelet — Grid, ichidagi qismlar — Flexbox

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Amaliy mashg'ulot

2-loyiha: portfolio sayti maketi

Amaliyot: header — Flexbox

CSS

```
* { box-sizing: border-box; margin: 0; padding: 0; }

.header {
  display: flex;
  justify-content: space-between; /* logo chapda, menyu o'ngda */
  align-items: center;
  padding: 16px 40px;
  box-shadow: 0 2px 10px rgba(0,0,0,.06);
}

.menu { display: flex; gap: 24px; }

.hero {
  min-height: 70vh;
  display: flex; flex-direction: column;
  justify-content: center; align-items: center;
  text-align: center;
}
```

Amaliyot: loyihalar — Grid

CSS

```
.loyihalar {  
  display: grid;  
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));  
  gap: 24px;  
  padding: 60px 40px;  
}  
.karta {  
  background: white;  
  border-radius: 14px;  
  box-shadow: 0 6px 20px rgba(0,0,0,.08);  
  overflow: hidden;  
}
```

- 1 "Men haqimda": rasm + matn — `display: flex; gap: 30px; align-items: center`
- 2 "Ko'nikmalar" nishonlari — `flex-wrap: wrap`
- 3 F12 → Elements: `grid` / `flex` belgisini bosing — brauzer to'rni chizadi

Ko'p uchraydigan xatolar

- ❌ `display: flex` ni **farzandga** yozish — u **otaga** yoziladi
- ❌ `justify-content: center` vertikal markazlamaydi — `align-items` kerak (va balandlik!)
- ❌ `box-sizing: border-box` unutilgan — elementlar qatorga sig'maydi
- ❌ `a`, `span` ga `width/height` — `inline-block` yoki `block` kerak
- ❌ `grid-template-columns: 1fr, 1fr` — vergul **qo'yilmaydi**

Savol

Element chegarasi bilan mazmuni orasidagi bo'shliq nima deyiladi?

A margin

B border

C padding

D gap

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

`width:200px; padding:10px; border:5px solid; + box-sizing: border-box .`

To'liq kenglik?

A 230px

B 200px

C 215px

D 210px

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Logotipni chapga, menyuni o'ngga surish uchun konteynerga nima yoziladi?

A `display:flex; justify-content: space-between`

B `display:flex; align-items: center`

C `display:grid; gap: 10px`

D `text-align: justify`

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

`grid-template-columns: 1fr 3fr;` — konteyner 800px (gap yo'q). Birinchi ustun necha piksel?

A 400px

B 100px

C 200px

D 267px

Javobni tanlang — to'g'ri variant yashil bo'ladi

Savol

Root shrift 16px bo'lsa, **2rem** necha piksel?

A 2px

B 16px

C 18px

D 32px

Javobni tanlang — to'g'ri variant yashil bo'ladi



Uyga vazifa

- 1 **Flexbox Froggy** (24 daraja) va **Grid Garden** (28 daraja) o'yinlari
- 2 Portfolio: "Men haqimda" (Flex), "Loyihalar" (Grid), "Ko'nikmalar" (`flex-wrap`), "Aloqa", footer
- 3 12 rasimli galereya: 2 tasi `span 2` bilan katta

✓ Yodda tuting

- Box model: content + padding + border + margin
- Har doim `* { box-sizing: border-box; }`
- Birliklar: `px`, `%`, `rem` (shrift), `vw/vh` (ekran)
- Flexbox: `justify-content` — asosiy o'q, `align-items` — kesishgan o'q, `gap`, `flex: 1`
- Grid: `repeat(auto-fill, minmax(200px, 1fr))`, `span`, `grid-template-areas`



Savollar bormi?

Keyingi mavzu: **9. CSS pseudo-class va pseudo-elementlar. CSS position**

Batafsil matn, misollar va testlar: **08-box-model-flexbox-grid.html** darsida